

2024, Mar. 2024 Release LOC:Acute Pediatric

Asthma

Utilization Benchmarks

Length of Stay: 1.6 d
 Length of Stay Type: InterQual
 Condition/Procedure: ASTHMA
 % Paid as Observation: 42%

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview

Select Day

Episode Day 1 (1, 2, 3)

Episode Day 2 (2, 3)

Episode Day 3

Discharge Screens (55)

Notes

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Overview

Introduction:

Asthma is a chronic inflammatory disorder of both the large and small airways causing recurrent episodes of wheezing, breathlessness, chest tightness, and coughing. These episodes are usually associated with widespread but variable airflow obstruction and may reverse spontaneously or with treatment. Asthma exacerbations are characterized by decreases in expiratory airflow that can be quantified and documented by measurement of peak expiratory flow (PEF) or forced expiratory volume in 1 second (FEV₁). These measurements, when obtainable, can be more accurate in reflecting the severity of the exacerbation than the symptoms being exhibited. Obtaining PEF or FEV₁ measurements in children may be developmentally difficult, particularly under the age of five.

Evaluation and Treatment:

The decision regarding appropriate disposition for a patient with asthma is based on the duration and severity of symptoms, severity of airflow obstruction, response to ED treatment or outpatient treatment, and risk factors for poor outcomes. Many clinical scoring systems such as Modified Pulmonary Index Score (MPIS), the Pediatric Asthma Severity Score (PASS), and the Pediatric Respiratory Assessment Measure (PRAM), have been validated for use in assessing the severity of asthma exacerbations in children and are recommended to assist in the decision to admit; however, the use of such scoring system is based on provider preference and hospital policy (Miller et al., Respir Care 2020, 65: 1227-32; Maue et al., World J Clin Pediatr 2017, 6: 34-9; Ducharme et al., J Pediatr 2008, 152: 476-80, 80 e1). The initial assessment of asthma exacerbation includes a history and physical examination, assessment of PEF or FEV₁ (if appropriate), determination of oxygen saturation, and other tests as indicated. Acute treatment may include (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Expert Panel Working Group of the National Heart et al., J Allergy Clin Immunol 2020, 146: 1217-70; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- Inhaled short-acting bronchodilators (SABA) and ipratropium
- Systemic corticosteroids
- Oxygen, noninvasive positive pressure ventilation (NIPPV), or mechanical ventilation for respiratory support
- Magnesium sulfate, heliox, and parenteral beta-agonists, such as terbutaline, may be administered as

adjunctive therapy in some cases

Many pediatric asthma patients are at high risk for readmission and careful attention to discharge planning is essential.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Expert Panel Working Group of the National Heart et al., J Allergy Clin Immunol 2020, 146: 1217-70; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882).

Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:** ^(1, 2, 3)

(Symptom or finding within 24h)

● **OBSERVATION, Both:** ⁽⁴⁾● Finding, ≥ **One:**● High risk for decompensation, ≥ **One:** ⁽⁵⁾● Barrier to discharge or arrangements for outpatient services ≤ 24h, ≥ **One:** ⁽⁶⁾

- Limited access to prompt medical care
- Parent or caregiver inability to provide care
- Outpatient services unavailable

- History of asthma requiring intubation and mechanical ventilation

- Hospitalization or ED visit for asthma within the past year

- Poor adherence with asthma medications and/or poor adherence with an asthma action plan

- Pregnancy

● Oral corticosteroid use, **One:**- Age ≤ 5 yrs, current or within the past year ⁽⁷⁾- Age > 5 yrs, current or within the past month ⁽⁸⁾- Comorbidity ^(9, 10)

- Use of > 1 canister of short-acting beta-agonist per month

● Partial response to at least 2h of initial treatment and, ≥ **One:** ^(11, 12, 13, 14, 15)

- Wheezing, persistent

- Inadequate PO intake ⁽¹⁶⁾- O₂ sat 92 - 95%(0.92 - 0.95) and < baseline ⁽¹⁷⁾- PEF or forced expiratory volume at one second (FEV₁) 40-69%(0.40-0.69) ^(18, 19)● Respiratory rate, **One:** ⁽²⁰⁾

- 50-59/min, sustained (age < 3 mos)

- 40-49/min, sustained (age 3 to < 18 mos)

- 30-39/min, sustained (age 18 mos to < 4 yrs)

- 20-29/min, sustained (age 4 to < 18 yrs)

● Intervention, ≥ **One:**- Corticosteroid (includes PO) ⁽²¹⁾- IV fluid ^(22, 23)- Short-acting beta-agonist ≥ 2 doses ^(24, 25)● **ACUTE, Both:** ⁽²⁶⁾● Partial response to at least 2h of initial treatment and, **One:** ^(11, 12, 13, 15, 27)● Wheezing, persistent and, ≥ **One:**- Nasal flaring, grunting, or retractions ⁽²⁸⁾- O₂ sat <92%(0.92) and < baseline ⁽²⁹⁾● Respiratory rate, **One:** ⁽³⁰⁾

- ≥ 60/min, sustained (age < 3mos)

- ≥ 50/min, sustained (age 3 to < 18 mos)

- ≥ 40/min, sustained (age 18 mos to < 4 yrs)

- ≥ 30/min, sustained (age 4 to < 18 yrs)

● Intervention, ≥ **One:**- Corticosteroid (includes PO) ^(12, 21)- FiO₂ > 21%(0.21) to < 35%(0.35) ^(31, 32)- Short-acting beta-agonist ≥ 6x/24h ^(24, 25, 33)● **INTERMEDIATE, ≥ One:** ⁽³⁴⁾● Chronic ventilation and, **One:** ⁽³⁵⁾● Mechanical ventilation and, ≥ **One:**- Increase in baseline FiO₂ > 21%(0.21) to 39%(0.39) ⁽³⁶⁾- Increase in baseline duration of ventilator use ⁽³⁶⁾

- NIPPV and, ≥ **One:** ⁽³⁷⁾
 - Increase in baseline $\text{FiO}_2 > 21\%(0.21)$ to $39\%(0.39)$ ⁽³⁶⁾
 - Increase in baseline duration of NIPPV use ⁽³⁶⁾
- HFNC and, ≥ **One:** ⁽³⁸⁾
 - Flow rate, **One:** ⁽³⁹⁾
 - $\leq 20 \text{ L/min}$ ($> 10 \text{ kg}$)
 - $< 2 \text{ L/kg/min}$ ($\leq 10 \text{ kg}$)
 - $\text{FiO}_2 > 21\%(0.21)$ to $39\%(0.39)$ ^(31, 32)
 - Oxygen $35\text{-}39\%(0.35\text{-}0.39)$ ^(31, 32)
 - Short-acting beta-agonist every 2h ⁽⁴⁰⁾
- CRITICAL, ≥ **One:** ⁽⁴¹⁾
 - Chronic ventilation and, **One:** ⁽³⁵⁾
 - Mechanical ventilation and, ≥ **One:**
 - Change in baseline setting ^(36, 42)
 - $\text{FiO}_2 \geq 40\%(0.40)$
 - NIPPV and, ≥ **One:** ⁽³⁷⁾
 - Change in baseline setting ^(36, 43)
 - $\text{FiO}_2 \geq 40\%(0.40)$
 - Heliox ⁽⁴⁴⁾
 - HFNC and, ≥ **One:** ⁽³⁸⁾
 - Flow rate, **One:** ⁽⁴⁵⁾
 - $> 20 \text{ L/min}$ ($> 10 \text{ kg}$)
 - $\geq 2 \text{ L/kg/min}$ ($\leq 10 \text{ kg}$)
 - $\text{FiO}_2 \geq 40\%(0.40)$ ⁽³²⁾
 - Mechanical ventilation
 - NIPPV ⁽³⁷⁾
 - Oxygen $\geq 40\%(0.40)$ ^(31, 32)
 - Short-acting beta-agonist every 1h or continuous ⁽⁴⁶⁾
 - Terbutaline ⁽⁴⁷⁾
- Episode Day 2, **One:** ^(2, 3)
 - OBSERVATION, **One:**
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽⁴⁸⁾
 - Wheezing, resolved or resolving, **All:**
 - O_2 sat within acceptable limits ⁽⁴⁹⁾
 - PEF or forced expiratory volume at one second (FEV_1) within acceptable limits or not indicated ^(49, 50)
 - Vital signs stable or within acceptable limits ⁽⁴⁹⁾
 - Caregiver able to manage patient at home and follow-up as needed
 - Tolerating PO ⁽⁵¹⁾
 - Partial responder, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Asthma exacerbation unresolved, **Both:**
 - Finding, ≥ **One:**
 - Wheezing, persistent
 - O_2 sat $92 - 95\%(0.92 - 0.95)$ and $< \text{baseline}$ ⁽¹⁷⁾
 - PEF or forced expiratory volume at one second (FEV_1) $40 - 69\%(0.40 - 0.69)$ ^(18, 19)
 - Respiratory rate, **One:** ⁽²⁰⁾
 - $50\text{-}59/\text{min}$, sustained (age < 3 mos)
 - $40\text{-}49/\text{min}$, sustained (age 3 to < 18 mos)
 - $30\text{-}39/\text{min}$, sustained (age 18 mos to < 4 yrs)
 - $20\text{-}29/\text{min}$, sustained (age 4 to < 18 yrs)
 - Short-acting beta-agonist ≥ 2 doses ⁽²⁵⁾
 - Inadequate oral intake and IV fluid ^(16, 22, 23)
 - ACUTE, **One:**
 - Responder, discharge expected today if clinically stable last 24h, **Both:** ⁽⁴⁸⁾

- Wheezing, resolved or resolving, **All:**
 - O₂ sat within acceptable limits ⁽⁴⁹⁾
 - PEF or forced expiratory volume at one second (FEV₁) within acceptable limits or not indicated ^(49, 50)
 - Vital signs stable or within acceptable limits ⁽⁴⁹⁾
 - Tolerating PO ⁽⁵¹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Pneumonia (**use Infection: Pneumonia subset**)
- Asthma exacerbation unresolved, **Both:**
 - Finding, ≥ **One:**
 - Wheezing, persistent
 - Nasal flaring, grunting, or retractions ⁽²⁸⁾
 - O₂ sat <92%(0.92) and < baseline ⁽²⁹⁾
 - Respiratory rate **One:** ⁽³⁰⁾
 - ≥ 60/min, sustained (age < 3mos)
 - ≥ 50/min, sustained (age 3 to < 18 mos)
 - ≥ 40/min, sustained (age 18 mos to < 4 yrs)
 - ≥ 30/min, sustained (age 4 to < 18 yrs)
 - Intervention, ≥ **One:**
 - FiO₂ > 21%(0.21) to < 35%(0.35) ^(31, 32)
 - Short-acting beta-agonist ≥ 6x/24h ^(25, 33)
 - Inadequate oral intake and IV fluid ^(16, 22, 23)
 - Post critical care ≤ 24h
- **INTERMEDIATE**, ≥ **One:**
 - Chronic ventilation and, ≥ **One:** ⁽³⁵⁾
 - FiO₂ > 21%(0.21) to 39%(0.39)
 - Respiratory intervention every 3-4h ⁽⁵²⁾
 - Weaning to baseline ≤ 2d ⁽³⁶⁾
 - HFNC and, ≥ **One:** ^(38, 53)
 - Flow rate, **One:** ⁽³⁹⁾
 - ≤ 20 L/min (> 10 kg)
 - < 2 L/kg/min (≤ 10 kg)
 - FiO₂ > 21%(0.21) to 39%(0.39) ^(31, 32)
 - NIPPV ⁽³⁷⁾
 - Oxygen 35-39%(0.35-0.39) ^(31, 32)
 - Short-acting beta-agonist every 2h ⁽⁴⁰⁾
- **CRITICAL**, ≥ **One:**
 - Chronic ventilation and, ≥ **One:** ⁽³⁵⁾
 - Change in baseline setting ^(36, 54)
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽⁵²⁾
 - Heliox ⁽⁴⁴⁾
 - HFNC and, ≥ **One:** ^(38, 53)
 - Flow rate, **One:** ⁽⁴⁵⁾
 - > 20 L/min (> 10 kg)
 - ≥ 2 L/kg/min (≤ 10 kg)
 - FiO₂ ≥ 40%(0.40) ^(31, 32)
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽³⁷⁾
 - FiO₂ ≥ 40%(0.40)
 - Respiratory intervention every 1-2h ⁽⁵²⁾
 - Oxygen ≥ 40%(0.40) ^(31, 32)
 - Short-acting beta-agonist every 1h or continuous ⁽⁴⁶⁾
 - Terbutaline ⁽⁴⁷⁾

● **Episode Day 3, One:**● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽⁴⁸⁾● **Wheezing, resolved or resolving, All:**– O₂ sat within acceptable limits ⁽⁴⁹⁾– PEF or forced expiratory volume at one second (FEV₁) within acceptable limits or not indicated ^(49, 50)– Vital signs stable or within acceptable limits ⁽⁴⁹⁾– Tolerating PO ⁽⁵¹⁾● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**

– For a condition other than asthma use appropriate subset (Episode Day 1) based on clinical findings

– Refer for secondary review

● **ACUTE, One:**● **Responder**, discharge expected today if clinically stable last 24h, **Both:** ⁽⁴⁸⁾● **Wheezing, resolved or resolving, All:**– O₂ sat within acceptable limits ⁽⁴⁹⁾– PEF or forced expiratory volume at one second (FEV₁) within acceptable limits or not indicated ^(49, 50)– Vital signs stable or within acceptable limits ⁽⁴⁹⁾– Tolerating PO ⁽⁵¹⁾● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**– **Use Extended Stay subset**

– Refer for secondary review

● **INTERMEDIATE, One:**● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**– **Use Extended Stay subset**

– Refer for secondary review

● **CRITICAL, One:**● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**– **Use Extended Stay subset**

– Refer for secondary review

Discharge, Level of Care, One: ⁽⁵⁵⁾● **HOME, All:**● **Level of care appropriateness, Both:**– Home environment safe and accessible ⁽⁵⁶⁾

– Patient or caregiver demonstrates ability to manage care

● **Asthma discharge planning, All:** ⁽⁵⁷⁾– Provide comprehensive written discharge and teaching instructions including asthma action plan ⁽⁵⁸⁾● **Delivery device, ≥ One:**

– Choose appropriate delivery device

– Assessment of delivery device technique for all medications

– Reinforcement of correct technique

– Review technique with PEF meter

– Consider referral for case or care management if poorly controlled or frequent visits to the ED

● **Education on the importance of, All:**

– Strict medication adherence

– Schedule follow-up care ^(59, 60)● **Precipitating factors, ≥ One:**

– Identify factors precipitating exacerbation

– Develop strategies to avoid precipitating factors ⁽⁶¹⁾

- Asthma self-management
- Provide a written asthma action plan or review the patients existing plan
- Instruction on recognition of signs of recurrence and worsening of asthma
- Influenza vaccine ⁽⁶²⁾
- Communication with school health personnel, if applicable ⁽⁶³⁾
- Medication reconciliation ⁽⁶⁴⁾
- Follow-up care planned
- Identify and address transportation needs
- **HOME CARE, All:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽⁵⁶⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(65, 66)
 - Skilled services, ≥ **One:** ⁽⁶⁷⁾
 - Skilled nursing ⁽⁶⁸⁾
 - Skilled therapy (PT, OT, or ST)
 - Behavioral health ⁽⁶⁹⁾
 - Asthma discharge planning, **All:** ⁽⁵⁷⁾
 - Provide comprehensive written discharge and teaching instructions including asthma action plan ⁽⁵⁸⁾
 - Delivery device, ≥ **One:**
 - Choose appropriate delivery device
 - Assessment of delivery device technique for all medications
 - Reinforcement of correct technique
 - Review technique with PEF meter
 - Consider referral for case or care management if poorly controlled or frequent visits to the ED
 - Education on the importance of, **All:**
 - Strict medication adherence
 - Schedule follow-up care ^(59, 60, 70)
 - Precipitating factors, ≥ **One:**
 - Identify factors precipitating exacerbation
 - Develop strategies to avoid precipitating factors ⁽⁶¹⁾
 - Asthma self-management
 - Provide a written asthma action plan or review the patients existing plan
 - Instruction on recognition of signs of recurrence and worsening of asthma
 - Influenza vaccine ⁽⁶²⁾
 - Communication with school health personnel, if applicable ⁽⁶³⁾
 - Medication reconciliation ⁽⁶⁴⁾
 - Follow-up care planned
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁷¹⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽⁷²⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁷³⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁶⁸⁾
 - Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(74, 75)
- Functional impairments requiring at least supervision ⁽⁷⁶⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁶⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- SUBACUTE MEDICAL OR THERAPY, **Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - Services, ≥ **One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁶⁸⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(74, 75)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁶⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁶⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- SUBACUTE REHAB, **Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(74, 75)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁶⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁶⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- ACUTE REHAB, **Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(74, 75)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁷⁶⁾
 - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁷⁷⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁶⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- LONG TERM ACUTE CARE, **Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁷⁸⁾

- Ventilator dependent $\geq 6\text{h/d}$ and weaning planned ^(79, 80)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁸¹⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁶⁴⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Notes:**1:**

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. Asthma patients at highest risk for readmission include those with a history of prior asthma admissions (Hogan et al., J Pediatr 2021, 236: 219-28.e11; Bloomberg et al., Am J Respir Crit Care Med 2003; 167(8): 1068-1076).

2:**Expected progress after 12-24h**

- Peak expiratory flow (PEF) or forced expiratory volume in one second (FEV₁) returning to baseline
- Vital signs normalizing (heart rate, respiration rate), no longer requires accessory muscle breathing
- Can speak in complete sentences without dyspnea, as age appropriate
- Weaning off O₂
- Decreased wheezing/dyspnea
- O₂ saturation improving

Consider:

- Decreasing frequency of nebulized treatments or metered-dose inhaler (MDI)
- Decreasing dose or frequency of corticosteroids
- Discharge planning

3:**Care Facilitation**

Potential risk for readmission

- Knowledge deficit or nonadherence with condition management identified (**Home Care, Readmission reduction program**)

Social Determinants of Health (SDOH)

Social Determinants of Health (SDOH) are conditions in a person's environment that may affect their health. Social needs related to SDOH may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. Children and caregivers should be screened for SDOH-related barriers to health to mitigate any unmet social and psychosocial factors impacting the health of the child and family unit (Kreuter et al., Annu Rev Public Health 2021, 42: 329-44; Sokol et al., Pediatrics 2019, 144:). The following social needs should be assessed on admission to determine which interventions are relevant to the individual child and their family support system. This list may not be all inclusive and all relevant SDOH factors should be considered. Any social need determined to be relevant to a given child and family should be planned for throughout the hospital stay.

*Interventions should be considered for all patients and families if social need is relevant

Alcohol or substance use

- Coordinate care with outpatient primary care or behavioral health providers*
- Refer to addiction social support systems*
- Provide information about community resources and programs for family-centered treatment (e.g., intensive case management, detox, outpatient, residential treatment, day treatment, and medication-assisted treatment)
- Provide information about home visitation programs (e.g., Parent-Child Assistance Program (PCAP) for pregnant and parenting individuals with substance use disorders aftercare) (Hildebrandt et al., Infant Ment Health J 2020, 41: 677-96; Vanderplasschen et al., Front Psychiatry 2019, 10: 186)
- Provide information about Cognitive Behavioral Therapy (CBT) Multidimensional Family Therapy (MDFT) and/or pharmacotherapy for youth with co-occurring emotional/mental disorders (Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021)

Child maltreatment, abuse, or intimate partner abuse - sexual and/or psychological abuse

- Arrange for alternative housing situation, foster care, or shelter if home environment is unsafe*
- Assist with creating a long-term safety plan or make appropriate referral*
- Refer to child protective services (CPS) and other domestic violence services if warranted*
- Refer for psychosocial support if warranted*
- Provide community resources and legal information

- Provide information on parent training programs such as the Nurse-Family Partnership Program (NFP) (Nurse-Family Partnership National Resources for Families. 2021; Gubbels et al., Int J Environ Res Public Health 2019, 16:)

End of life planning, arrange palliative care or hospice

- Consider palliative care for children with end stage disease
- Consider hospice care for terminally ill children with a life expectancy of less than 6 months

Environmental exposure to toxic substances and other physical hazards

Lead poisoning:

- Provide community resources for primary prevention and secondary prevention strategies (e.g., blood lead level testing and follow up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Tobacco use (e.g., cigarettes, e-cigarettes, vaping):

- Provide education and/or brief counseling about tobacco use cessation and the risk of second-hand smoke (U.S. Department of Health and Human Services, Tobacco Use in Children and Adolescents: Primary Care Interventions. 2021)

Financial resources strained or limited

Food insecurity:

- Provide information about local food banks and special nutritional programs (e.g., Special Supplemental Nutrition Program for Women, Infants, and Children (WIC), U.S. Department of Agriculture Summer Food Service Program) (Testa and Jackson, Ann Epidemiol 2021, 58: 22-8; Berman et al., Pediatr Rev 2018, 39: 235-46)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

Income insecurity:

- Identify income resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to direct income assistance benefits and indirect income assistance programs (e.g., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)

Infants and children with Special Health Care Needs

- Provide anticipatory guidance and education to caregivers regarding medical condition and expected progress after discharge over the next few days, months, or years, as warranted (Dosman and Andrews, Paediatr Child Health 2012, 17: 75-80)*
- Refer for follow-up care to neonatologist or other medical subspecialist for infants or children with ongoing and/or chronic medical issues (e.g., bronchopulmonary dysplasia or feeding dysfunction) (Pediatrics 2008, 122: 1119-26)*
- Refer for adaptive/assistive medical equipment and equipment for technology-dependent children (e.g., home oxygen, ventilators, monitors, or tube feeding) (Pediatrics 2008, 122: 1119-26)*
- Refer for neurodevelopmental assessment for high-risk infants*
- Refer to post-discharge follow-up programs (e.g., Maternal, Infant, and Early Childhood Home Visiting Program, Early Head Start program for infants or toddlers under the age of 3, and pregnant individuals in need of medical, mental health, nutrition and education, or the Nurse-Family Partnership for mothers pregnant with their first child and for ongoing nurse care visits through the child's second birthday) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Refer for behavioral support and/or counseling for caregivers through therapy groups, telephone support, and/or peer support groups for caregivers of children with similar health needs (Reyes et al., Health Equity 2021, 5: 100-18; Substance Abuse and Mental Health Administration, Treatment Considerations for Youth and Young Adults with Serious Emotional Disturbances and Serious Mental Illnesses and Co-occurring Substance Use. 2021; Pierron et al., BMC Public Health 2018, 18: 1087)
- Refer to lactation consultant in the hospital and home visits for infants at risk for poor lactation or suboptimal infant breastfeeding behavior (Leeman et al., Pediatr Qual Saf 2019, 4: e130)
- Refer for respite services, if warranted

Mental health co-morbidities involved

- Arrange appointment with primary care provider or behavioral health specialist upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information about suicide prevention, including the National Suicide Prevention hotline*
- Refer for psychiatric services for suicide risk screening and intervention
- Refer to school-based mental health interventions, if applicable (Fazel et al., Lancet Psychiatry 2014, 1: 377-87)
- Provide information about Intensive Community-Based treatment (e.g., Functional Family Therapy, Intensive Child and Adolescent and Psychiatric Services, Multidimensional Family Therapy, and Intensive Case Management)
- Transfer to day treatment program, residential addiction services, or partial hospitalization when warranted

Risk for treatment nonadherence

- Provide information about educational and behavioral support initiatives for children at risk of medication nonadherence and their caregivers, such as text message reminders to take medication and electronic medication dispensers
- Provide information about peer support groups for children and/or families with poor insight into illness or noncompliance with treatment (Mental Health America, Peer Support: Research and Reports 2021; National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- Refer to interventions to prevent medical relapse or readmission and to enhance medication compliance

Risk for readmission

- Consider transfer to an alternative level of care such as subacute care
- Consider home care services
- Consider early childhood home visiting programs (e.g., Early Head Start; Early On; Nurse Family Partnership; and the Maternal, Infant, and Early Childhood Home Visiting Program) (Health Resources & Services Administration, Maternal and Child, The Maternal, Infant, and Early Childhood Home Visiting Program 2021; Nurse-Family Partnership National Resources for Families. 2021)
- Consider telehealth and web-based training programs (e.g., telepsychiatry or telebehavioral health and mobile health interventions) (J Am Acad Child Adolesc Psychiatry 2017, 56: 875-93)
- Consider intensive in-home behavioral health services
- Provide patient and family with condition-specific information including appropriate planning for prevention of respiratory syncytial virus (RSV) infection and appropriate immunizations

Special Health Risk Behaviors in adolescence and young adulthood

- Arrange follow-up appointment with primary care provider and/or specialists upon discharge*
- Coordinate care with outpatient behavioral health providers*
- Provide information on mentoring programs (e.g., Big Brothers Big Sisters of America)
- Refer for Intensive Community-Based treatment for children with comorbid behavioral health issues
- Refer to community resources for teen pregnancy prevention/family planning (e.g., use of birth control, subsequent pregnancies) (Maness et al., J Adolesc Health 2016, 58: 636-43)

For eating disorders:

- Provide information on nutrition services including education and resources for diet management, physical activity, and promotion of healthy eating attitudes and behaviors (Gato-Moreno et al., Int J Environ Res Public Health 2021, 18;; Al-Khudairy et al., Cochrane Database Syst Rev 2017, 6: Cd012691)
- Provide information on outpatient psychosocial interventions and Family-Based Treatment (Lock and La Via, J Am Acad Child Adolesc Psychiatry 2015, 54: 412-25)

For parenting resources:

- Provide information on Family Resource Centers and Parenting support groups for young, first-time parents (e.g., National Parent Helpline) (National Parent and Youth Helpline)
- Provide information on community resources to improve parenting skills (e.g., face-to-face, by telephone) (U.S. Department of Health and Human Services, Parenting Resources. 2021)

Unstable living environment or homelessness expected upon discharge

- Assist caregivers with health insurance application
- Provide information about homeless shelter application to caregivers if applicable
- Refer to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- Refer to federally sponsored outpatient programs
- Refer to Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community Treatment (Ponka et al., PLoS One 2020, 15: e0230896; Vijverberg et al., BMC Psychiatry 2017, 17: 284)

4:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

5:

These risk factors have been shown to increase the risk of asthma-related mortality and asthma severity (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Pena et al., Emerg Med Clin North Am 2017, 35: 625-45; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007). Patients who respond to initial treatment and have any of these risk factors may require continued observation for risk of decompensation, to address adherence issues, and to ensure a support system is in place prior to discharge.

6:

There is limited evidence to recommend absolute barriers to discharge for pediatric patients. Literature does support that children with limited access to prompt medical care, especially children who do not have a primary care provider, are at increased risk of readmission and noncompliance with prescription regimens. Children who come from limited-income households or have a parent or caregiver who is physically or mentally unable to care for them are also at higher risk of not getting the necessary follow-up care if discharged from the hospital without social service involvement (Johns Hopkins University and Quality, Improving the Emergency Department Discharge Process: Environmental Scan Report. 2014).

7:

Current oral steroid use is a marker of asthma severity and may indicate an increased risk for asthma-related mortality. A course of oral steroids within the past year due to a severe asthma exacerbation has been linked to poor outcomes in patients 5 years of age or younger (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023). These patients may require continued monitoring at the Observation level of care.

8:

Current oral steroid use is a marker of asthma severity and may indicate an increased risk for asthma-related mortality (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023). These patients may require continued monitoring at the Observation level of care. Although the literature does not define a specific time for "recently stopped" oral steroid use in patients greater than 5 years of age, the definition of "within the past month" is based upon best practice as determined by InterQual's® external peer reviewers. This determination is based on an iterative process involving multiple clinicians with diverse expertise in varied practices and geographic settings.

9:

Comorbidities include:

- Cardiovascular disease
- Chronic lung disease in addition to asthma

- Chronic psychiatric condition

10:

Most available literature and data focus on the adult population, but guidelines list cardiovascular disease, chronic lung disease other than asthma, and chronic psychiatric conditions as comorbidities associated with higher risk for asthma-related death (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

11:

Initial treatment for an acute asthma exacerbation consists of inhaled short-acting beta-agonists and systemic corticosteroids. Epinephrine (for anaphylaxis), inhaled ipratropium bromide, and/or intravenous magnesium may also be used in some cases (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

12:

Instruction: Treatment is inclusive of interventions provided in a medical practitioner's office, outpatient clinic (e.g., urgent care center), or emergency department.

13:

Treatment with both a short-acting beta-agonist and ipratropium bromide were associated with fewer hospitalizations and greater improvement in peak expiratory flow and forced expiratory volume than a short-acting beta-agonist alone in adults and children with a severe asthma exacerbation. With repeated doses, however, the benefits of ipratropium decline. Hospitalized children who continued to receive ipratropium as adjunctive therapy did not experience additional benefits or shorter hospital stays (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Leung, Drugs Context 2021, 10; Griffiths and Ducharme, Cochrane Database of Systematic Reviews 2013:).

14:

In patients who present with acute asthma exacerbation, frequent reassessments after the initial management with short-acting beta-agonist (SABA) and titration of further treatments based on the clinical response are recommended. If peak expiratory flow or forced expiratory volume in 1 second is being used to evaluate the patient, the recommendation is to repeat the measurement 1 hour after the initial treatment to assess the response to bronchodilator treatments (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

15:

Initial treatment for an acute asthma exacerbation includes:

- At least 2 doses of short-acting beta-agonists
- Ipratropium bromide (unless not indicated)
- Systemic corticosteroids (including PO) given in last 24 hours

16:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

17:

Mild to moderate asthma exacerbation may be associated with oxygen saturation between 90-95%(0.90-0.95); however, in younger children (i.e. 5 years of age or younger), oxygen saturation of less than 92%(0.92) could indicate severe asthma exacerbation (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management

and Prevention. 2023).

18:

Forced expiratory volume at 1 second (FEV₁) is the maximum amount of air that can be forcefully exhaled in 1 second. It is then converted to a percentage predicted. FEV₁ is a marker for the degree of obstruction caused by asthma. Although peak expiratory flow (PEF) may not be as reliable as FEV₁, PEF is another way to measure degree of obstruction and can be helpful in evaluating severity of asthma and response to asthma treatments. Patients, with education, can perform and monitor PEF with a peak flow meter at home and PEF is often included as a part of asthma action plan. There are predicted FEV₁ and PEF values available based on height, weight, sex assigned at birth, and/or race; however, a patient's own personal best value, if available, is recommended for use when trying to evaluate the severity of asthma exacerbation (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007). FEV₁ and PEF measurements can be used to assess the severity of asthma exacerbation as follows (National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- $\geq 70\%$ (0.70) predicted/personal best = Mild exacerbation
- 40% to 69%(0.40 to 0.69) predicted/personal best = Moderate exacerbation
- $< 40\%$ (0.40) predicted/personal best = Severe exacerbation

These measurements can be more reliable indicators of severity than symptoms if able to be performed correctly. However, measuring FEV₁ or PEF may not be possible in children in distress due to acute asthma, and many children under 5 years of age are unable to perform these tests properly. If FEV₁ or PEF is being used to evaluate the patient, the recommendation is to repeat the measurement 1 hour after the initial treatment (and at intervals) to assess the response to bronchodilator treatments (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Leung, Drugs Context 2021, 10:).

19:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

20:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

21:

Patients who have received corticosteroids within the past 24 hours will meet this line of criteria. Corticosteroids decrease airway inflammation and early administration is associated with reduced ED visits, hospitalization, and

duration of symptoms. Oral steroids can usually be used as they are usually quicker and less invasive, but intravenous dosing may be preferred when there are concerns for vomiting or patients are too dyspneic. Frequency and duration differ depending on the type of corticosteroid as follows (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Leung, Drugs Context 2021, 10:):

- Prednisolone/prednisone - 1 -2 mg/kg/day for 3-5 days
- Dexamethasone - 0.15-0.6mg/kg/dose. One dose within 1 day or 2 doses 24 hours apart may be given to the patient

22:

To allow for the advancement of feedings as intravenous (IV) fluids are weaned, the IV fluid rates are $\geq 50\%$ (0.50) of the maintenance fluid requirement. These rates can be calculated using the following (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020):

Weight ≤ 10 kg = ≥ 50 mL/kg/24h

- Hourly rate = 2.08 (mL/h) x kg

Weight > 10 -25 kg = ≥ 40 mL/kg/24h

- Hourly rate = 1.67 (mL/h) x kg

Weight > 25 -60 kg = ≥ 30 mL/kg/24h

- Hourly rate = 1.25 (mL/h) x kg

Example: An 11 kg child is required to receive a minimum of 40 mL/kg/24h to meet this criteria point. Multiply 1.67 (mL/h) x 11 (child's weight in kg) = 18.37 mL/h. This is the minimum fluid replacement rate for this child.

23:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

24:

Instruction: Short-acting beta-agonist doses refer to the total number of doses given regardless of frequency of treatments. Doses here do not include ones given prior to decision to admit.

25:

Instruction: This criterion applies to the administration of a bronchodilator via nebulizer or metered-dose inhalers (MDI) with or without a spacer. MDIs produce outcomes that are at least equivalent to nebulizers. The use of a spacer may improve medication delivery via an MDI for patients who have difficulty with proper technique. Spacers may have some advantages compared to nebulizers for children with acute asthma. The use of an inhaler with a spacer is also recommended when viral transmission via nebulizer can lead to increased risk of COVID-19 (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007; Cates et al., Cochrane Database Syst Rev 2013: CD000052).

26:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

27:

Frequent reassessments after the initial management with short-acting beta-agonist (SABA) and titration of further treatments based on the clinical response are recommended. If lung function is being used to evaluate the patient, the recommendation is to repeat the measurement 1 hour after the initial treatment to assess the response to bronchodilator treatments (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and

Prevention. 2023).

28:

Airway obstruction in children with asthma may manifest as the excessive use of accessory muscles of respiration such as the scalene, intercostal, and diaphragmatic, which are indicative of respiratory distress and the need for admission (Miller et al., Respir Care 2020, 65: 1227-32). Retractions, including intercostal, subcostal, substernal, supraclavicular, and especially deep suprasternal, are also signs of moderate to severe respiratory distress in children (Maue et al., World J Clin Pediatr 2017, 6: 34-9; Ducharme et al., J Pediatr 2008, 152: 476-80, 80 e1).

29:

Different international and national guidelines recommend different thresholds of pulse oximetry values for oxygen supplementation. Saturation less than 92%(0.92) has been associated with high morbidity and the need for hospitalization (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

30:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxia, ventricular arrhythmia, or hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., the resolution of hypertension following anti-hypertensive therapy or the resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® expert clinical consultants. The criteria are based upon current best practice and are the product of an iterative process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

31:

Guidelines recommend supplemental oxygen use in acute asthma exacerbation. Target oxygen saturation levels differ between guidelines, but the latest recommendations from Global Initiative of Asthma (GINA) in 2023 suggest keeping oxygen saturation at 93%(0.93) to 95%(0.95) for those older than 11 years old and at 94%(0.94) to 98%(0.98) for 11 years old and younger (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Paton, British guideline on the management of asthma. SIGN 158. 2019; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

32:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO_2] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO_2) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO_2 optimally delivered:

Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(0.24-0.35) FIO_2

Reservoir systems (For neonates, infants, and young children, the maximum FIO_2 achievable is not well documented)

- Simple oxygen masks 35-50%(0.35-0.50) FIO_2
- Partial-rebreathing masks 40-60%(0.40-0.60) FIO_2
- Non-rebreathing masks > 60%(0.60) FIO_2

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(0.24-0.40) FIO_2

- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(0.21-1.0) FIO₂

Enclosure systems

- Oxygen hoods / tents 21-100%(0.21-1.0) FIO₂

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

33:

Inpatient management of asthma relies on ongoing reassessments and adjustment of treatment plan. Many hospitals use asthma pathways that guide the adjustment of the frequency of short-acting beta-agonist (SABA) treatments throughout the day depending on patient's response. SABA treatments may be spaced out to less frequent dosing if the patient shows signs of improvement. If there is worsening of respiratory status after weaning, the patients may be moved back to the previous, more frequent SABA treatments (Ewulonu and Dyer, Curr Probl Pediatr Adolesc Health Care 2021, 51: 100995).

34:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

35:

Chronic ventilation refers to a patient who is managed at home on chronic mechanical ventilation via a tracheostomy tube or noninvasive positive pressure ventilation via a mask. This includes continuous positive airway pressure (CPAP) or bilevel positive airway pressure (BiPAP).

36:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

37:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

38:

High flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO₂) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates set independent from the FiO₂. The heated and humidified gas has been shown to improve comfort, aid in the clearance of secretions, and reduce bronchospasms. When flow

rates are set above the patient's inspiratory demand, the upper airway is washed out of carbon dioxide, nasal resistance is decreased, and there is a reduction in dead space, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. Bronchiolitis is the main indication for HFNC in pediatric patients beyond the neonatal period. Although the benefits of HFNC in other conditions such as pneumonia, asthma, and post-extubation have been reported, more well-designed studies are needed to guide the use of HFNC in these conditions (Kwon, Clin Exp Pediatr 2020, 63: 3-7).

39:

High flow nasal cannula (HFNC) consists of three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of the other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). Standard HFNC flow rates in pediatrics have yet to be established (Kwon, Clin Exp Pediatr 2020, 63: 3-7). Although most studies on HFNC in pediatrics focus on children with bronchiolitis, they can provide some guidance on appropriate flow rates outside of the intensive care unit (ICU). Initial flow rates in clinical trials have found that 1-2 L/kg/min is effective in patients with bronchiolitis (Yurtseven et al., Pediatr Pulmonol 2019, 54: 894-900). However, a recent survey revealed that only 19%(0.19) of responders use a weight-based flow rate method (Miller et al., Respir Care 2018, 63: 894-9). This method may not be optimal in older children as it can result in very high flow rates that may not be tolerated. Some studies combat this potential problem by setting maximum flow rates. Flow rates up to 20 L/min are well tolerated on non-ICU wards (Willer et al., Hosp Pediatr 2021, 11: 891-5; Kepreotes et al., Lancet 2017, 389: 930-9). Therefore, a flow rate of 20 L/min was chosen as the maximum for the Intermediate level of care.

Instruction: To calculate the L/kg/min, the flow rate is divided by the patient's weight.

Example:

HFNC flow rate: 12 L/min

Patient weight: 8 kg

$12/8 = 1.5$

This patient is on a flow rate of 1.5 L/kg/min

40:

Instruction: patient can meet this line of criteria if unable to be weaned from every 2 hour treatment within 12 hours.

Inpatient management of asthma relies on ongoing reassessments and adjustment of treatment plan. Many hospitals use asthma pathways that guide the adjustment of the frequency of short-acting beta-agonist (SABA) treatments throughout the day depending on patient's response. SABA treatments may be spaced out to less frequent dosing if patient improves. If there is worsening of respiratory status after weaning, patients may be moved back to the previous, more frequent SABA treatments (Ewulonu and Dyer, Curr Probl Pediatr Adolesc Health Care 2021, 51: 100995).

41:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

42:

A change in 1 or more of the following settings from baseline would meet at the Critical level of care:

- Inspiratory time (iTime)
- Positive end-expiratory pressure (PEEP)
- Pressure support (PS)
- Respiratory rate
- Tidal volume (TV)

43:

A change in 1 or more of the following settings from baseline would meet at the Critical level of care:

- Inspiratory positive airway pressure (IPAP)

- Expiratory positive airway pressure (EPAP)
- Positive end expiratory pressure (PEEP)

44:

Heliox, a combination of helium and oxygen administered through an oxygen delivery system, has been used in the treatment of severe asthma. Helium improves the flow of oxygen through restricted airways due to its lower density. Its use is not routinely recommended but may be considered for patients with refractory symptoms (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Craig et al., Cochrane Database Syst Rev 2020, 8: CD012977).

45:

High flow nasal cannula (HFNC) consists of the following three parameters: temperature, fraction of inspired oxygen (FiO_2), and flow rate. Each parameter is independent of each other. The flow rate is how fast the heated and humidified air/oxygen is delivered per minute and is documented as liters per minute (L/min). The flow rate is often based on the patient's weight in smaller children (Kwon, Clin Exp Pediatr 2020, 63: 3-7). To calculate the liters per kilogram per minute (L/kg/min), the flow rate is divided by the patient's weight.

Example:

HFNC flow rate: 12 L/min

Patient weight: 8 kg

$12/8 = 1.5$

This patient is on a flow rate of 1.5 L/kg/min

46:

Instruction: This criterion includes administration of aerosolized medication via nebulizer or inhaler. Appropriate selection of a delivery device for aerosol therapy depends on multiple factors including cognitive state or developmental stage, manual dexterity, breathing pattern, respiratory drive, and drug formulation. Efficacy is optimized when the device is used correctly and complements the patient's age, physical characteristics, and cognitive ability (Tashkin, Int J Chron Obstruct Pulmon Dis 2016, 11: 2585-96; Brittan et al., J Pediatr 2015, 166: 998-1005 e1).

47:

Terbutaline is a systemic beta-agonist that can be used in severe acute asthma exacerbation. Evidence behind the effectiveness or benefit of terbutaline is limited, but it continues to be mentioned as a possible treatment choice by guidelines. When given intravenously, a bolus is usually followed by an infusion that requires close monitoring of vital signs and adverse effects while adjusting the rate according to clinical status of the patient. Adverse effects can include tachycardia, hypotension, and hypokalemia (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; Leung et al., Drugs Context 2021, 10;; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007). InterQual's® external peer reviewers agree that terbutaline is a treatment utilized at the Critical level of care.

48:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

49:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

50:

Recommendations differ on the acceptable threshold for peak expiratory flow (PEF) or forced expiratory volume at 1 second (FEV_1) values for safe discharge. Guidelines from the National Heart, Lung, and Blood Institute state that post-treatment PEF or FEV_1 of greater than or equal to 70%(0.70) is generally an appropriate cutoff to discharge patients from the ED or hospital. Patients with post-treatment PEF or FEV_1 less than that are recommended to be assessed individually for possible discharge based on risk factors. Global Initiative for Asthma (GINA) classifies

post-treatment PEF or FEV₁ of greater than 60-80%(0.60-0.80) as a moderate exacerbation and recommends considering discharging patients from the emergency department (ED) if post-treatment PEF or FEV₁ is greater than or equal to 60%(0.60) after considering risk factors and follow-up availability (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

51:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg/d*) can be calculated based on current body weight (*kg*):

- 100 *mL/kg/d* for first 10 *kg*
- 50 *mL/kg/d* for second 10 *kg*
- 20 *mL/kg/d* for each additional *kg* in weight

52:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

53:

Currently, there is no universally accepted method for weaning pediatric patients off high-flow nasal cannula (HFNC), and weaning protocols vary based on institutional policies. However, one commonly used method is to gradually decrease the fraction of inspired oxygen (FiO₂) to a specific level, such as 40%(0.40), then gradually reduce the flow rate before transitioning to a low-flow oxygen delivery system (e.g., nasal cannula). Less common methods include weaning the FiO₂ to a predetermined value, then transitioning to a low-flow oxygen system without reducing the flow rate, or gradually reducing the flow rate while keeping the FiO₂ constant (Kawaguchi et al., *Pediatr Crit Care Med* 2020, 21: e228-e35). There is no standard flow rate at which HFNC should be discontinued, but based on the literature, it is suggested to discontinue HFNC when the flow rate reaches 2 L/min in infants, and 4 L/min in children older than 12 months (Udurgucu et al., *Pediatr Allergy Immunol Pulmonol* 2022, 35: 79-85; Peterson et al., *Respir Care* 2021, 66: 591-9).

Instruction: When weaning a patient off high-flow nasal cannula (HFNC), it is expected that the current level of care be maintained or a lower level of care from the previous level be utilized to conduct the review if HFNC criteria are used. A typical weaning strategy usually does not involve simultaneously decreasing the flow rate while increasing the FiO₂. This scenario may indicate that a patient is being kept on the HFNC machine but may be able to tolerate settings that could be administered by nasal cannula. For instance, a patient who was previously on HFNC 10 L/min, 30%(0.30) FiO₂ might be weaned to 3 L/min, but FiO₂ is increased to 100%(1.0). In this case, the patient should be assessed based on the flow rate rather than the FiO₂.

54:

A change in 1 or more of the following settings from baseline would meet at the Critical level of care:

- Expiratory positive airway pressure (EPAP)
- Inspiratory positive airway pressure (IPAP)
- Inspiratory time (iTime)
- Positive end expiratory pressure (PEEP)
- Pressure support (PS)
- Respiratory rate (RR)
- Tidal volume (TV)

55:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

56:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the

community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

57:

Evidenced-based guidelines recommend that patients utilize a daily diary and perform a periodic self-assessment (can be completed in the practitioner's office) to promote self-management skills and to identify their personal perception of asthma control. Patients should follow either a peak flow monitoring-based action plan and/or a symptom monitoring-based plan. Both have been found to be equally beneficial when taught and followed correctly by the patient. Furthermore, the self-monitoring plan must be individualized and tailored to meet patient and/or family preferences (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007).

58:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

59:

Patients should be referred to an asthma specialist in the following situations (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023; National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- When there has been a life-threatening asthma exacerbation
- If the goals of asthma therapy are not being met after 3 to 6 months of treatment
- When there are atypical signs and symptoms or difficulties confirming diagnosis of asthma
- If other conditions complicate asthma
- When additional education and guidance is needed (e.g., complications of therapy, problems with adherence, allergen avoidance)
- For consideration of immunotherapy
- If the patient has persistent asthma despite 2 or more medications
- If more than 2 courses of oral systemic corticosteroids are required in 1 year
- If there is a history of an occupational or environmental inhalant or ingested substance contributing to asthma

60:

The first follow up appointment is recommended within the first 1-2 days after hospital or emergency department discharge. An appointment with an asthma specialist should also be scheduled when indicated (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

61:

There are many triggers or precipitating events leading up to an asthma exacerbation. It is important to control or avoid those triggers whenever possible. Some suggestions to assist patients in this include (National Heart, Lung, and Blood Institute. Expert Panel Report 3: Guidelines for the Diagnosis and Management of Asthma - Full Report 2007):

- Smoking cessation
- Not smoking in the home or around children
- Encasing your mattress and pillows with dust-mite proof covers
- Decreasing indoor humidity below 60%(0.60)
- Keeping pet fur or hair out of the home

- Keeping all food out of the bedroom
- Keeping food and garbage in closed containers
- Using a vacuum with a high-efficiency particulate air (HEPA) filter or a double-layered bag
- Fixing leaky faucets, pipes, and other water sources
- Cleaning moldy surfaces
- Keeping windows closed during allergy season
- Asking your doctor when to increase medications
- Avoiding wood burning stoves
- Asking your doctor about taking medications before exercise to prevent symptoms
- Avoiding sulfites in food
- Covering your nose and mouth on cold windy days

62:

Patients with asthma who are being discharged should have their age-appropriate vaccination status reviewed. It is recommended that patients with asthma stay up-to-date with at least influenza and COVID-19 vaccinations if there are no contraindications to these vaccines (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

63:

For school-aged children, information shared by the hospital with school health personnel improves the patient's asthma control and decreases their primary care and emergency room visits (Harris et al., Cochrane Database Syst Rev 2019; 1: CD011651; Kew et al., Cochrane Database Syst Rev 2017; 4: CD012255).

64:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

65:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home

would be cumbersome and places the patient at risk

- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

66:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

67:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

68:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

69:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the

need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

70:

For children 5 years and younger, it is recommended that a regular assessment be conducted every 3 to 6 months to determine the need for continued asthma treatment (Global Initiative for Asthma (GINA), Global Strategy for Asthma Management and Prevention. 2023).

71:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

72:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

73:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

74:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

75:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

76:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

77:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

78:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

79:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

80:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

81:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.